

Analyzing and Visualizing the Correlation between Ecosystems and Environmental Sustainability : Focusing on Search API Data

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Abstract

This study aims to analyze public awareness of environmental sustainability, particularly focusing on the ecological importance of bees and their impact on human food production. Real-time search query data were collected and analyzed to identify trends related to bees and environmental keywords. Artificial intelligence techniques and data visualization methods were employed to create new forms of media art. The results of the study emphasize the importance of bees and can be used as a tool to promote a shift in public awareness regarding environmental issues.

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1 Introduction

Environmental issues are increasingly critical in modern society, with the decline in bee populations posing a direct threat to ecosystem stability and human food production. Bees play a crucial role in ecosystems, with 33% of human food production relying on their pollination activities. Despite this, public concern over the decline in bee populations remains low. This study seeks to analyze public awareness of bees and environmental sustainability by using real-time search query data and proposes a method for visualizing this data. The goal is to foster a shift in public awareness about environmental issues. proposes a method for visualizing this data. The goal is to foster a shift in public awareness regarding environmental issues.

2 DATA VISUALIZATION

2.1 Training Dataset

In this study, images related to bees were utilized as the training dataset for AI. To this end, bee disease diagnosis image data were collected from AI Hub, and images of healthy specimens were selected to construct a dataset for use in style transfer techniques.

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This dataset is designed to visually emphasize the ecological role of bees and comprises over 30,000 images. It provides a foundation for effectively visualizing the importance of bees.

2.2 Naver DATA LAB

This study utilized search query APIs, such as Naver Data Lab, to collect search query data. The collected data includes real-time search queries related to bees, environmental sustainability, and crops produced through bee pollination.

Data Collection: Real-time search data for keywords related to bees, environmental sustainability, and bee-pollinated products were obtained through search query APIs. This data provides essential information for analyzing public interest and awareness.

Data Preprocessing: The collected data undergoes preprocessing to prepare it for analysis. During this process, relevant keywords and search patterns are extracted and formatted into suitable graphical representations for analysis.

Trend Analysis: Based on the preprocessed search data, public interest in bees and environmental sustainability is assessed. Analysis includes examining search frequency, trend changes, and seasonal factors to understand how public interest evolves over time.

2.3 Style Transfer

Style Transfer is a technique that applies the style of one image to another using deep learning. In this study, images of healthy bees collected from AI Hub were used as content images. Graphical images reflecting the search keywords related to environmental sustainability were set as style transfer images. This approach was employed to generate new visuals that reflect the visual changes corresponding to the keywords.

2.4 TouchDesigner

TouchDesigner is a tool suitable for creating real-time interactive content. In this study, it was utilized to visualize data collected from the search query API in real time. The tool's node-based interface allows for intuitive control of complex data flows and visual effects, and Python scripting enables advanced data processing and visualization. This facilitated the creation of media art that effectively communicates the importance of environmental issues to the public.

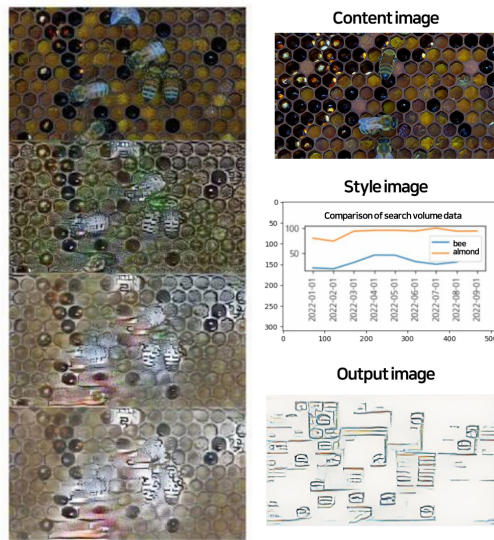


Figure 1: Style Transfer in Honeybee Image

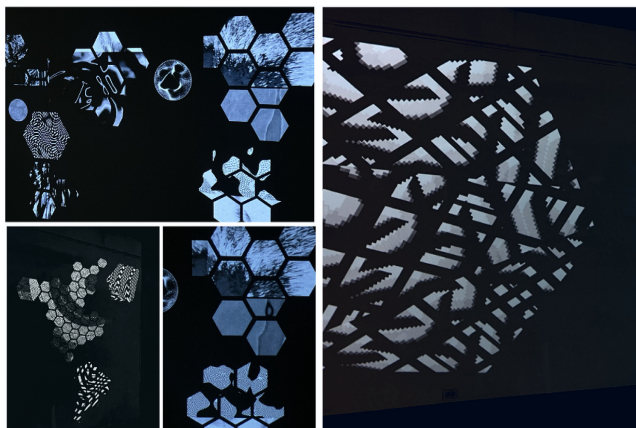


Figure 2: Results of Data Visualization and Media Art Transformation with Style Transfer

3 CONCLUSION

This study demonstrated the potential of using real-time search query data to analyze and visualize public awareness of environmental sustainability, with a specific focus on the role of bees in ecosystems. By integrating AI techniques, such as style transfer, with data visualization tools like TouchDesigner, we were able to create dynamic and visually compelling media art that effectively communicated the critical importance of bees. The results of this study highlight the underappreciated role of bees in food production and the broader ecosystem, emphasizing the need for increased public awareness and action to protect these vital pollinators. The integration of data analysis with media art offers a novel approach

to engaging the public and encouraging a shift in perception regarding environmental issues.

4 FUTURE RESEARCH

Future research will seek to expand the scope of this study by including a wider range of environmental issues, such as ocean pollution, deforestation, and the effects of climate change on various species. Long-term data collection and analysis will be conducted to capture seasonal trends, the impact of policy changes, and the influence of significant environmental events on public interest. Additionally, we plan to develop educational programs and campaigns based on the research findings, using data visualization and interactive media art to promote environmental awareness and encourage sustainable practices. Further, there will be an exploration of more advanced AI techniques and visualization methods to enhance the effectiveness of these tools in communicating complex environmental issues to the public.

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